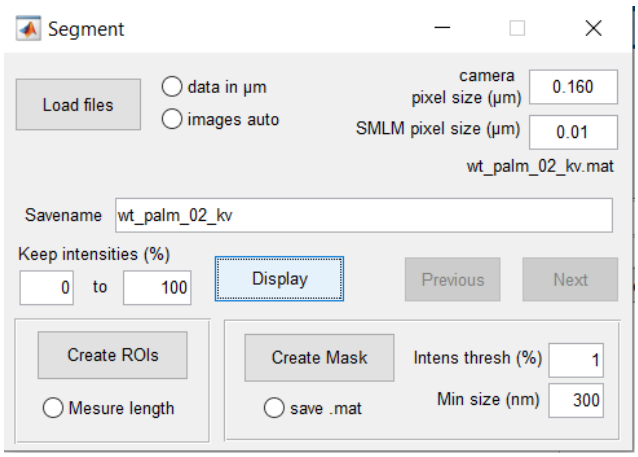
Segment

Graphic User Interface to select ROIs and to create segmented images from SMLM data.

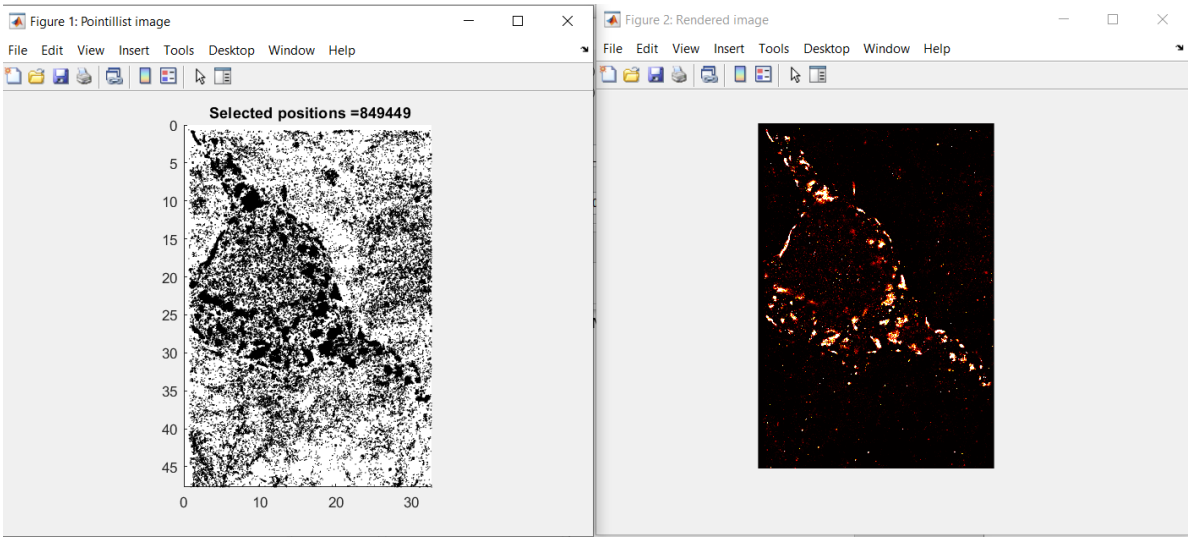


Enter the **camera pixel size** and the **SMLM pixel size** (half of the localization precision) of your set-up.

Use **Load files** to select the files to be treated (.mat files). <u>If your data is already calibrated in microns, click on **data in µm** before loading the files.</u>

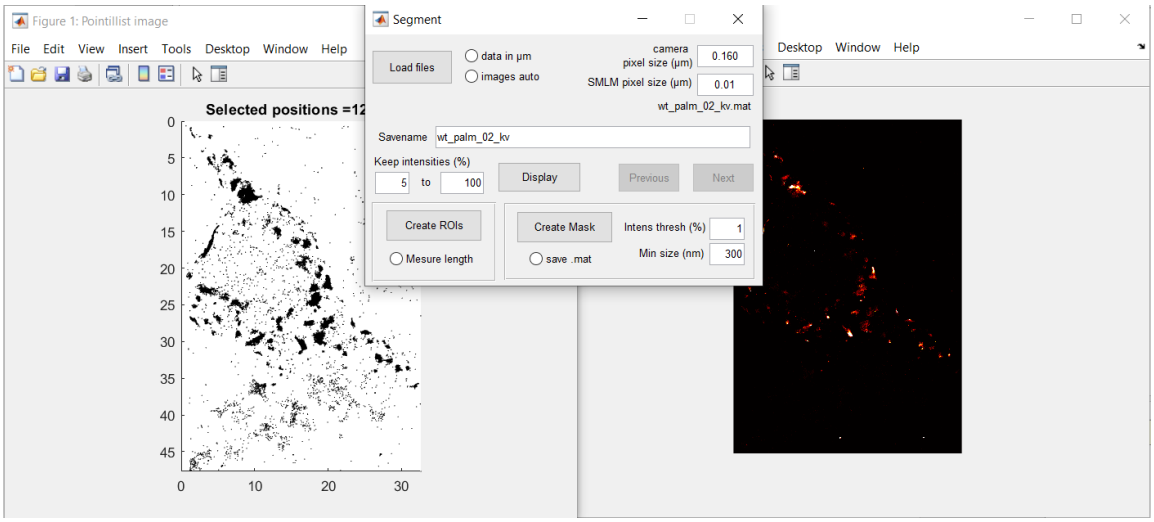
The resulting images will be saved with the **Savename** indicated (modifiable). Be careful if ROIs are going to be used for clustering analysis, as they have to be automatically recognized by the analysis program. The name has to be *name*.mat for the detection file, and *name*.rgn for the ROI datafile.

If **images auto** is selected, the pointillistic and rendered images will be displayed automatically for the active file. Otherwise, you can use **Display** to visualize them. If you loaded a list of files, use **Next** or **Previous** to move to the next or previous file, respectively.



Keep intensities (%) can be useful to eliminate low intensity or high intensity spurious signals: set the low and high thresholds (expressed as percentage of maximum intensity) and use Display to verify the selection.

For the previous example:



<u>Please note: the ROI (.rnd) will be created from the *original* set of detections, whereas the mask file will be created using the detections whose intensity is within the low and high limits</u>.

Use **Create ROIs** to select ROIs by hand and to manage previous selections. If you want to add a distance measurement, select **Mesure length**.

Create Mask will create a segmented image from the rendered image (no ROI in this case).

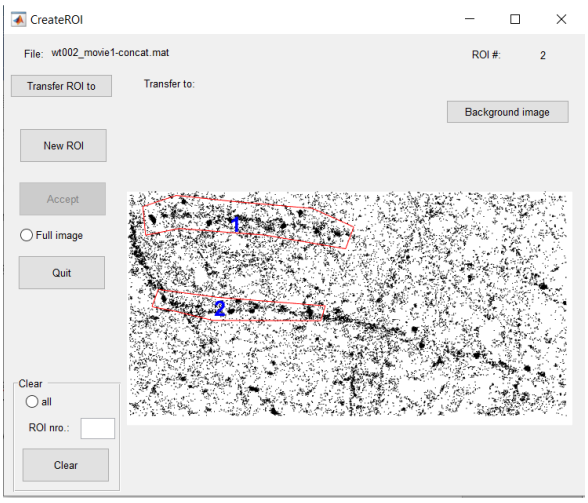
Create ROIs

This will open a new window where you can select ROIs by hand on the pointillistic image. If there is already a selection for the active file (a .rgn file), this selection will be shown.

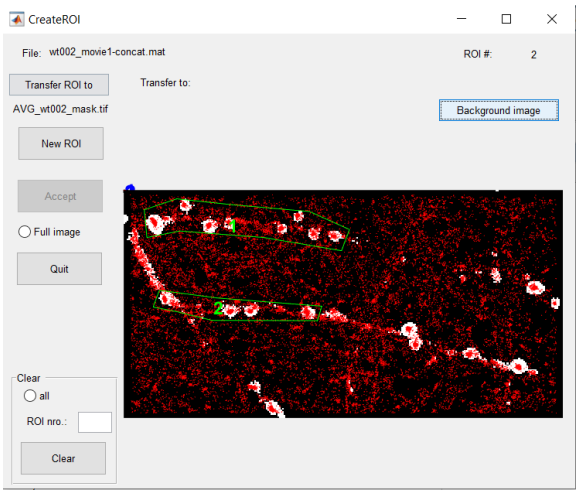
You can overlay a background image (.tif) loading it with **Background image**. **Transfer ROI to** will create the same ROIs also in the selected .mat file (please note that both images must have the same size). **Quit** to come back to the Segment window.

Example with previous ROI data (2 ROIs):

No background:



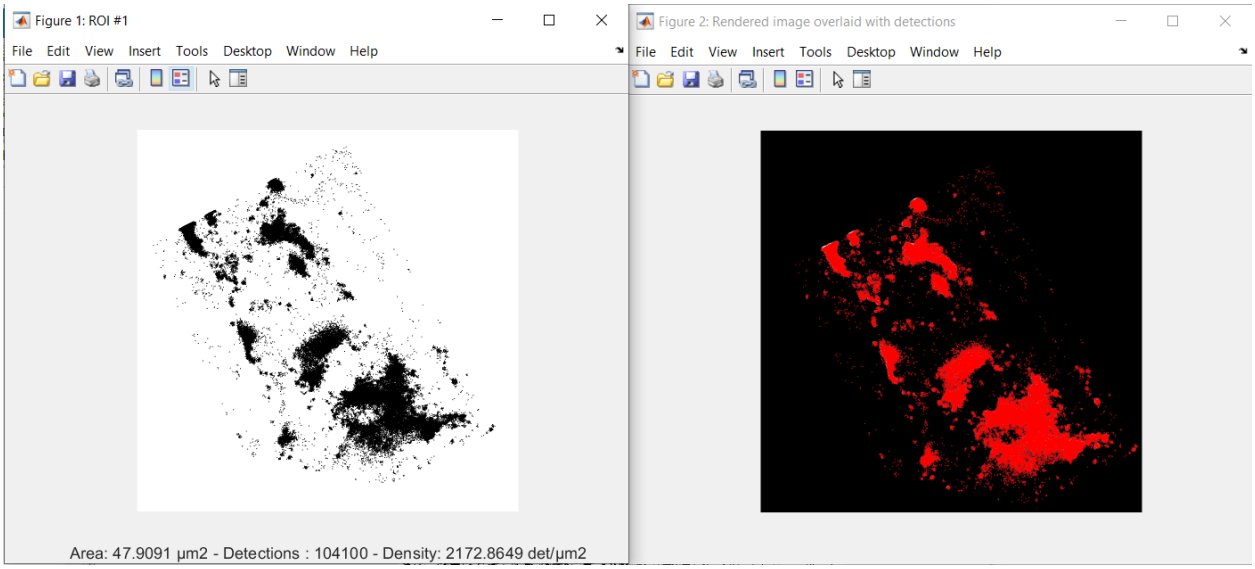
With background:



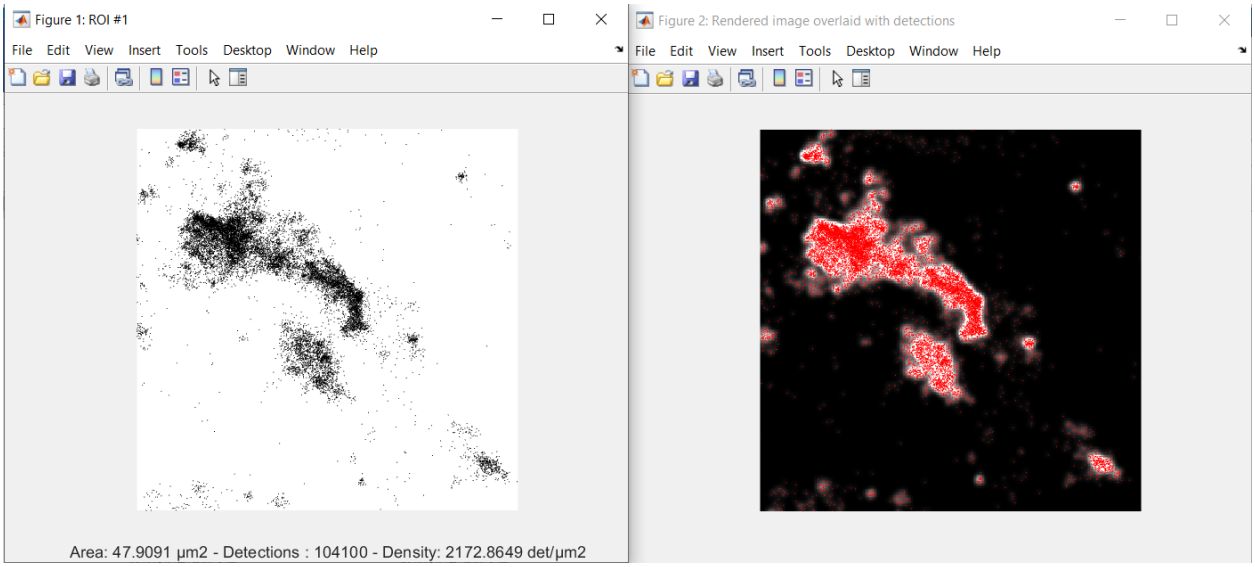
Drawing ROIs

You can select a polygon on the image (ROI) by clicking on **New ROI**. If you want to keep the whole image as ROI, choose **Full image** before clicking on New ROI.

After drawing the polygon (double click to finish it), the pointillistic and the rendered image of the ROI (overlaid with the detections in red) will pop-up.



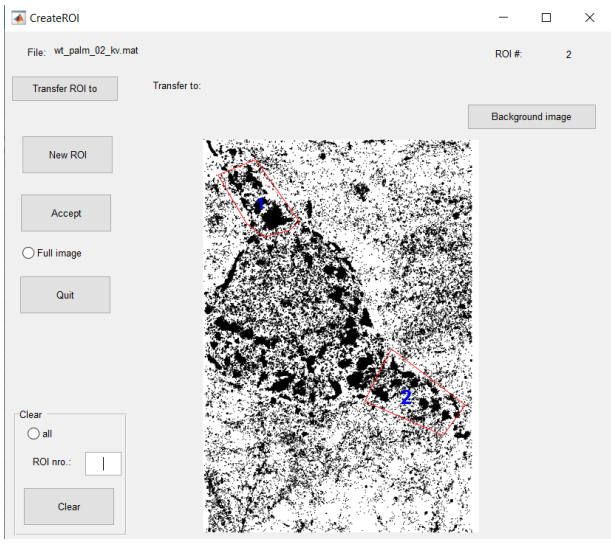
A zoom on the images:



Note that on the pointillistic image window you have the area of the ROI, the number of detections found inside and the density of detections in the ROI, assuming a homogeneous distribution.

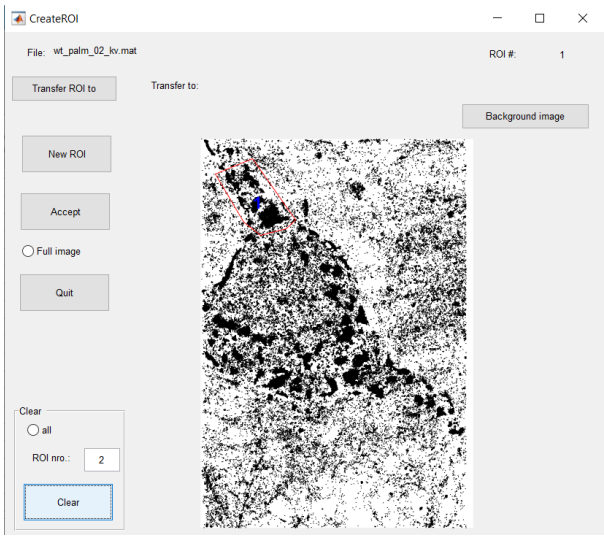
On the CreateROI window, click on **Accept** to keep the selection, or click on **New ROI** if you want to select another one. “Accept” will number the ROI and show this number on the window.

You can add as many ROIs as you want. As an advice to facilitate clustering analysis, select rather small regions with a homogeneous background level, if possible.



Deleting ROIs

You can delete one ROI using **ROI nro** and **Clear**. After deleting one ROI, the remaining ones are re-numbered. Choose **all** to delete all the ROIs.



The results will be saved in:

1) *name*-ROIinfo.txt
   Text file with a table where the columns are

ROI #	surface area (μm^2)	# detections in the ROI	density of the ROI (detections/μm^2)

2) *name*.rgn

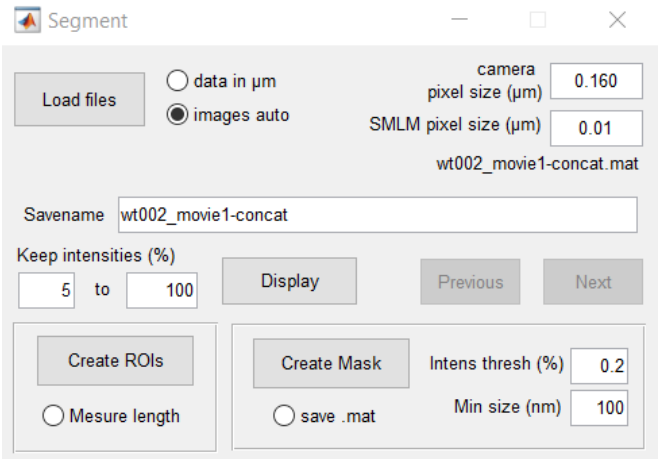
   .mat file with the structure ROI that contains the coordinates of the ROI and the detections that colocalize, among other data.

3) *name-roi#*-Irend.mat

   .mat file with the rendered image of the ROI, which is used by Cluster.m for clustering analysis.

Create Mask

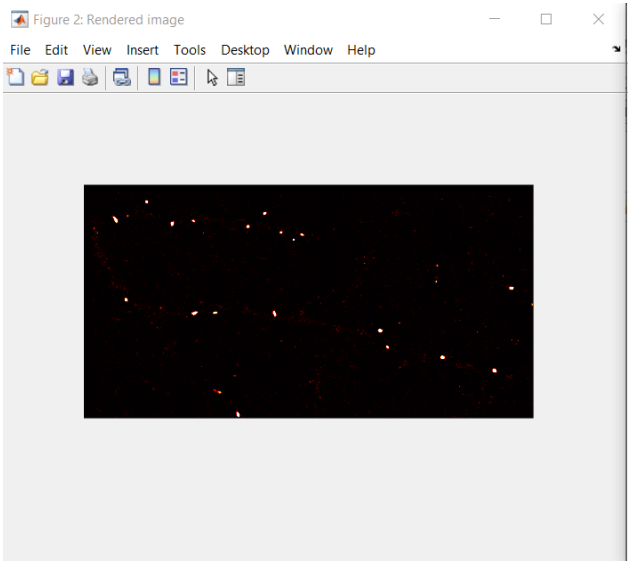
Create Mask makes a segmentation on the rendered image. You can set an intensity threshold with **Intens thresh** (in % of the maximum) and a minimum object size with **Min size**. This will select objects in the segmented image that have at least this size, assuming a circular shape. Play also with **Keep intensities** values to improve the segmentation.



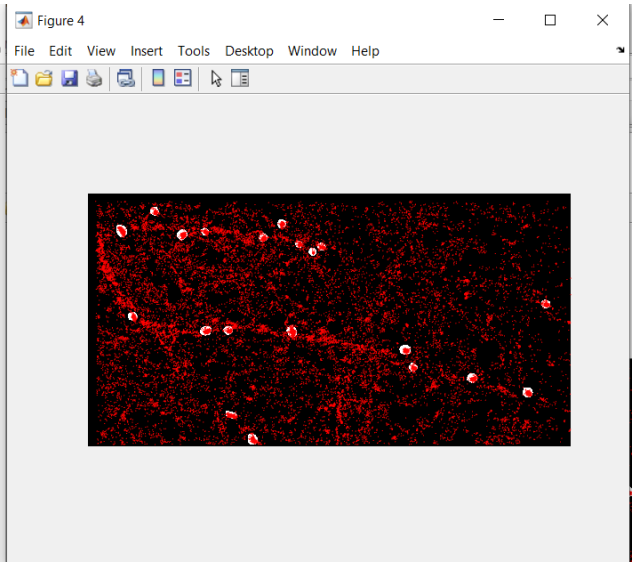
After clicking on **Create Mask**, a new window shows the result overlaid with detections plotted in red: zoom to see the objects in the mask that appear in white.

In comparison with the rendered image:

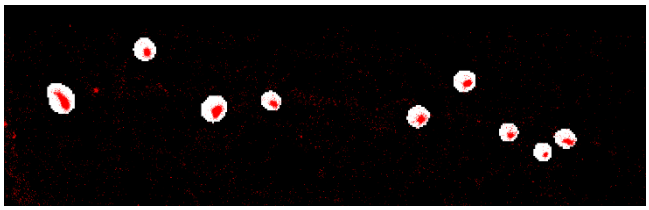
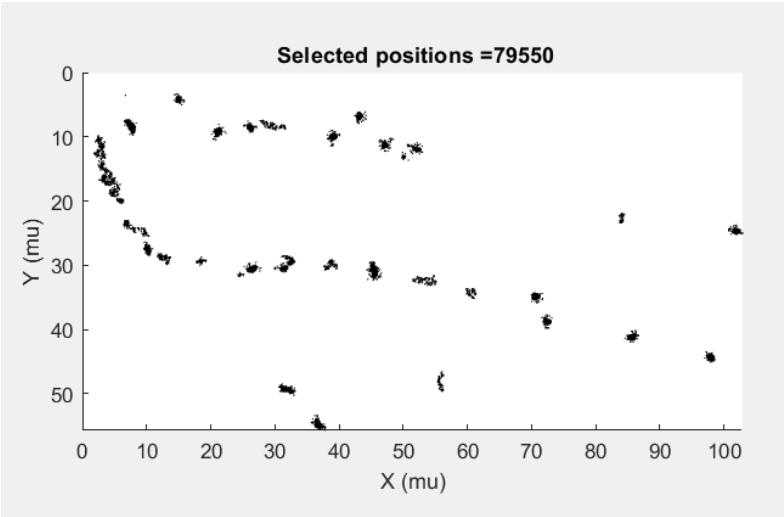
Rendered image



Mask overlaid with detections:



-> a detail after zooming:



Select **save .mat** to obtain a new .mat file that contains only detections in the selected objects (*name*-mask.mat). In the previous example, the pointillistic image of the resulting *name*-mask.mat file is: